Task 4 EEG waveform decoding

Response diagnostics, feature audits, and follow-up tests

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26 October 2026 | Analyses completed 21–24 October 2026

Main finding. Responses common to the five waveforms were more repeatable than waveform-specific differences. The tested voltage, xDAWN, alpha/beta-power and combined representations did not establish reliable overall five-class decoding. Expanded group response tests also found no corrected waveform, position or interaction effect. Conditional sensitivity checks recovered stronger injected reference patterns but missed all 0.25–1 μV cases; participant latency variation reduced recovery. These negative findings do not establish that waveform information is absent.

Question and scope

Given one trigger-aligned EEG epoch, can we identify which of five median-nerve TENS waveforms was delivered? The target is *delivered waveform identity*, not naturalness preference. Motivation comes from R. Collu et al.'s findings on waveform-dependent sensation [1]; perceptual differences do not by themselves imply a separable EEG signature. This note extends the initial Task 4 baseline shared previously. No Task 3 decoding result is included.

Following initial event, signal-quality and response checks, the work prioritized representation and decoding while retaining a targeted preprocessing comparison. It covers four areas:

- **Response characterization:** electrode/position diagnostics, local EEG slopes, multiscale representations and time-resolved group tests.
- **Preprocessing assessment:** matched minimal-versus-supplied processing, pre-trigger controls and contamination indicators.
- **Feature assessment and decoding:** repeated numerical audits and fixed xDAWN/power comparisons with the original voltage reference.
- **Uncertainty and sensitivity:** participant-bootstrap intervals and recovery of known injected reference patterns.

Shared data context

All analyses use ten current-protocol participants (subjects 3–12). Subject 2 had an incompatible Task 4 version. Jason Whitson's corrected mapping is Sine=239, Square=240, Triangle=241, linear decrease=242 and linear increase=243. Task 4 is a pairwise naturalness task with a race-to-five stopping rule; choices are subjective, not correct/incorrect responses. The thesis provides the original experimental context [2]; current labels and protocol clarifications follow Jason's correspondence.

Analysis track	Events	Complete pairs	Electrodes
Voltage / xDAWN / power classifiers	1,244	622	14 predefined
Response diagnostics and refinements	1,242	621	All 62 scalp

All 1,246 debounced events matched the full behavioral sequence. The two tracks apply different fixed EEG quality/support checks, so their retained cohorts must not be interchanged. Recorded sampling is 1 kHz. Within-pair intervals were 1.408–1.448 s; reported post-response windows end by 500 ms. Physical trigger-to-current delay remains unverified.

1 EEG response characterization

1.1 Electrode responses, presentation position and local slopes

The 62-channel descriptive track adapts the supplied processing: 0.5-Hz high-pass, 60-Hz harmonic notches, average reference, 33-component extended Infomax ICA fitted at full rate on a 1-Hz high-pass copy, automatic Fp1-based EOG exclusion, interpolation from -2 to $+10$ ms, and 190-Hz low-pass. Original fits were reused for subsequent analyses without refitting. Full-recording ICA is descriptive here; it is not claimed as a fitted held-out classifier transform.

Fixed windows summarized voltage, signed slope and absolute per-trial slope at each electrode; slopes used a 25-ms local-linear fit. Windows were pre $[-200, -50)$, early $[50, 150)$, middle $[150, 300)$ and late $[300, 500)$ ms. Participant regressions included comparison intercepts, centered elapsed time, waveform, first/second position and their interaction.

Repeated complete-pair splits distinguished the response shared across waveforms from *waveform deviations*, obtained by subtracting the equally weighted common response. These are group medians of participant median split-half correlations, not independent split estimates:

Representation, 50–500 ms	Position	Common r	Waveform-deviation r
Voltage	First	0.386	0.020
Voltage	Second	0.686	0.078
Local slope	First	0.176	0.006
Local slope	Second	0.207	0.023

A repeatable common response can coexist with weak waveform separation. No corrected waveform, position or interaction effect was found (Table 2). Non-significant position tests do not establish equivalence or eliminate carryover, expectation or recording-order confounds.

1.2 Multiscale representations and participant uncertainty

Gaussian-smoothed voltage and first derivatives at $\sigma = 5, 10, 20$ ms were compared with original voltage and local slopes; difference-of-Gaussians (DoG, 5–10 and 10–20 ms) were secondary contrasts. Gaussian derivatives were unit-ramp calibrated in μV/s; DoG remained in μV. All representations used the same 100 whole-pair splits and 100–500-ms trace support. Repeatability used 599 pairs; 22 singleton ordered-comparison cells were omitted only from this check, not from the 621-pair response models.

Participant median correlations were paired across representations, then the ten participants were resampled 10,000 times. Table 1 gives median paired changes from the local slope. Intervals are descriptive pointwise 95% percentile intervals, without multiplicity adjustment.

Gaussian derivative	Position	Median Δr	Descriptive 95% interval
$\sigma = 5$ ms	First	$+0.005$	$[-0.001, +0.010]$
$\sigma = 5$ ms	Second	-0.004	$[-0.012, +0.001]$
$\sigma = 10$ ms	First	-0.018	$[-0.046, +0.018]$
$\sigma = 10$ ms	Second	$+0.011$	$[-0.002, +0.042]$
$\sigma = 20$ ms	First	-0.032	$[-0.089, +0.048]$
$\sigma = 20$ ms	Second	$+0.033$	$[-0.015, +0.103]$

Table 1. Waveform-repeatability changes relative to the 25-ms local slope. Participant identity is preserved in every paired resample.

All six intervals span zero. Smoothed-voltage and DoG waveform-repeatability changes also had intervals spanning zero. The 20-ms derivative's paired *common-response* changes were +0.137 (first; interval 0.040–0.187) and +0.261 (second; 0.129–0.358). Smoothing emphasized common structure rather than a demonstrated discriminative signature. These pointwise intervals do not determine joint usefulness for single-epoch decoding or establish neural origin.

1.3 Time-resolved effects and earlier coverage

Window averages can hide brief or sign-changing effects. A separate test fit the participant regression at every electrode/time point for Gaussian derivatives at $\sigma = 2.5, 5, 10, 20, 40$ ms in both processing branches. Four sum-coded waveform coefficients and four interaction coefficients were tested using the sum of their squared group t statistics; position used one coefficient. This is a model-conditional omnibus statistic, not a parametric F-distribution test.

All 1,024 participant sign patterns were enumerated, applying each participant's sign jointly across the full contrast vector and map. Independent participants and joint symmetry of coefficient errors under the null are required. Individual trial labels were not shuffled. Maximum-statistic correction spanned the entire declared location family; a further factor of three covered waveform, position and interaction. These locations are correlated, not independent Bonferroni tests.

Response test	Locations/family	Waveform	Position	Interaction
Electrode / local-slope windows	744	0.7910	0.0820	1.0000
Multiscale mean/RMS windows	4,960	1.0000	1.0000	0.1582
Time-resolved, common late support	372,000	1.0000	1.0000	0.8848
Time-resolved, expanded support	492,776	1.0000	0.6973	1.0000

Table 2. Minimum corrected p in each effect family. No location reached corrected $p \leq .05$. Families differ across experiments; changes in their minima are not tests of differences between experiments. Mean/RMS denotes signed mean and per-trial root-mean-square.

The original time-resolved analysis used post $[180, 480)$ and pre $[-480, -180)$ ms, accommodating the widest kernel. A subsequent exploratory extension, specified after the earlier negative result, allowed smaller kernels to inspect earlier times without selecting a peak:

Gaussian σ	Post interval (ms)	Pre interval (ms)	Samples/period
2.5 ms	$[50, 480)$	$[-480, -50)$	430
5 ms	$[50, 480)$	$[-480, -50)$	430
10 ms	$[51, 480)$	$[-480, -51)$	429
20 ms	$[91, 480)$	$[-480, -91)$	389
40 ms	$[171, 480)$	$[-480, -171)$	309

Each local kernel uses real recorded padding and avoids the explicit onset interpolation. The pre/post intervals match in duration and mirror the continuous half-open limits; they are not one fixed translation. Both branches, both periods, all scales, all valid times and all 62 electrodes remain in the expanded correction family. No correction was relaxed. Earlier continuous filters can still spread stimulation contamination.

For efficiency, voltage coefficients were estimated before the fixed linear Gaussian derivative: these two linear operations commute. Tests confirmed equivalence to trial-level filtering, and every previous late-window coefficient was reproduced with maximum absolute difference 1.08×10^{-11} in derived coefficient units. This extension is not an independently preregistered replication. A group null may miss participant- specific or latency-variable effects; a post-only significant result would not by itself establish a post-minus-pre difference.

2 Preprocessing assessment

2.1 Matched processing bundles

The targeted comparison asked whether the supplied processing could be suppressing repeatable waveform-specific patterns. Both branches used the same 1,242 events, 62-channel average reference and fixed representation operators, with no branch-specific rejection.

- **Minimal:** recorded EEG with verified units, average reference and interval-specific constant baseline subtraction; no added high-pass, notch, ICA, interpolation or low-pass.
- **Supplied:** the filtering/ICA/interpolation bundle described in Section 1, reconstructed using verified saved fits without refitting.

Acquisition-system filtering may already exist in the recorded input. The cohort was selected by the supplied-track quality rule; conclusions are conditional on this retained cohort rather than all raw trials.

The post trace $[100, 500)$ ms used baseline $[-250, -50)$ ms. The matched pre trace $[-500, -100)$ ms used an independent earlier baseline $[-850, -650)$ ms. Both trace and baseline are exact 600-ms translations. This corrects the voltage-baseline overlap present in the preserved original multiscale pre window. Constant baselines cancel for derivatives and DoG.

2.2 Repeatability and contamination indicators

Derivative	Position	Minimal: deviation r	Supplied: deviation r
$\sigma = 5$ ms	First	0.013	0.011
$\sigma = 5$ ms	Second	0.033	0.010
$\sigma = 10$ ms	First	−0.006	−0.003
$\sigma = 10$ ms	Second	0.052	0.013
$\sigma = 20$ ms	First	−0.033	0.006
$\sigma = 20$ ms	Second	0.041	0.044

Common responses were more repeatable than waveform deviations in both branches. Minimal processing gave small numerical increases for some second-presentation derivatives but no consistent advantage across scales and positions.

Exact participant sign flips jointly corrected 60 repeatability contrasts: post minimal-minus-supplied, pre minimal-minus-supplied, and their post-minus-pre difference, across ten representations and two positions. None survived correction; the minimum corrected p values for these three contrast groups were 1.0000, 0.6289 and 0.8359.

Median event-wise maximum-channel post peak-to-peak amplitude was 47.3 μV for minimal and 40.1 μV for supplied processing. No post event exceeded the diagnostic 500-μV flag in either branch; pre counts were three and one, respectively. Relative 60-Hz-band power was higher in minimal data. These are contamination indicators, not neural signal-to-noise estimates, and no flags changed eligibility.

Three of 60 new pointwise median-difference bootstrap intervals excluded zero, for second-presentation voltage/5/10-ms smoothed-voltage post-minus-pre bundle differences. They do not replace the earlier globally corrected mean-contrast tests, and no settings were selected from them.

Supported interpretation. There is no clear evidence here that the supplied bundle hid repeatable waveform-specific slopes at the tested scales. This is not equivalence, step-level validation, proof of artifact removal or a basis to choose a winning pipeline. Stage ablations were not performed; filtering before interpolation and full-recording ICA remain interpretation limits.

3 Feature assessment and waveform decoding

3.1 Repeated feature audits

The classifier track retained the original 1,244-event cohort, 14 predefined electrodes and minimal processing. The original audit covered five representations (post, pre, late, onset and order/history), totaling 300 training-fold audits. New representations were audited again after extraction: 120 fold-specific projected audits for xDAWN and 360 audits for the voltage/power/combined comparison, spanning pre/post and both validation schemes.

Checks covered finiteness, constant/near-constant columns, physical units, training-only scaling, held-out standardized extremes, correlations and within-class covariance rank/conditioning. Class-specific automatic Ledoit–Wolf covariance reconstructions matched the actual shrinkage-LDA covariances. Mixed μV and dB units were kept separate for near-constant review. No audit-driven clipping, exclusions or feature selection occurred.

All EEG feature arrays were finite with no constant/near-constant columns; some intentionally binary order/history controls were constant. All regularized classifier covariances had full numerical rank. The primary post-trigger diagnostics were:

Representation	Features	Empirical deficient[†]	Within cond.	LOSO cond.
Voltage	126	50/50	90.9	140.2
xDAWN	90	20/50	34.7	237.4
Alpha/beta power	28	0/50	22.4	261.5
Voltage + power	154	50/50	80.7	155.0

[†]Within-participant training folds. Condition numbers are medians of regularized covariance conditions; they are diagnostics, not universal pass/fail criteria.

For 126 voltage columns and 88–116 within-participant training epochs, empirical within-class rank deficiency is expected: rank $\leq \min(p, n_{\text{train}} - 5)$. Median entropy effective rank was 9.7/126 within and 15.3/126 in LOSO; these are not counts of informative neural dimensions. Fewer features improved some within-participant geometry without a uniform LOSO benefit. Large held-out excursions remained: post voltage reached $|z| = 16.9$ within and 37.0 LOSO; xDAWN 15.9 and 26.5; power 5.3 and 4.5.

Ten whole-pair covariance resamples per fold were conditional on the outer training scaler; xDAWN resamples additionally held the learned spatial filters fixed. Shrinkage resolved rank deficiency but did not uniformly reduce relative resampling changes around each estimator's own reference. These compact dependent resamples are not confidence intervals for the population covariance or the full spatial-learning procedure.

3.2 Fixed representations

All classifiers used training-only standardization and equal-prior shrinkage LDA (`lsqr`, automatic shrinkage), without a parameter search.

- **Voltage:** nine 50-ms means on 14 fixed electrodes, 126 columns.
- **xDAWN:** full-rate training epochs/labels only, two components per each of five classes, fixed 10% spatial-covariance shrinkage; ten mixtures with nine means give 90 columns. Separate pre/post fits and every permutation refit the spatial transform.
- **Power:** one full-epoch Hann periodogram after constant detrending, no padding; alpha 8–13 and beta 13–30 Hz integrals, expressed as $10 \log_{10}(P/1\,\mu\text{V}^2)$, give 28 columns. Native spacing is 2.22 Hz. Integrals use interpolated exact band boundaries. Voltage-plus-power has 154 columns. This is total single-trial power, including evoked and ongoing activity, not isolated induced oscillations. No class-specific ERP subtraction or learned spectral selection was used.

3.3 Validation, controls and results

Post epochs were $[50, 500)$ ms with voltage baseline $[-250, -50)$ ms; pre controls were $[-500, -50)$ ms with independent baseline $[-750, -550)$ ms. Constant detrending cancels voltage-baseline offsets for power; these are absolute epoch powers, not pre/post ratios. The minimal classifier path adds no temporal filter, ICA or onset interpolation.

Five within-participant folds held out complete waveform-comparison contexts, preserving pairs and purging shared temporal support. LOSO held out a complete participant. Every epoch received one held-out prediction. Balanced accuracy pooled the five-class out-of-fold predictions; within-participant means then weighted the ten participants equally. This tests context/participant generalization and is stricter than a random split of nearby trials.

Features	Within post	Within pre	LOSO post	LOSO pre
Voltage	19.05	16.75	18.52	19.35
xDAWN	20.70	17.79	20.03	20.25
Alpha/beta power	15.80	19.14	17.70	18.48
Voltage + power	19.09	17.21	18.12	19.79

All entries are balanced-accuracy percentages; 20% is a reference, not a substitute for a structured conditional null.

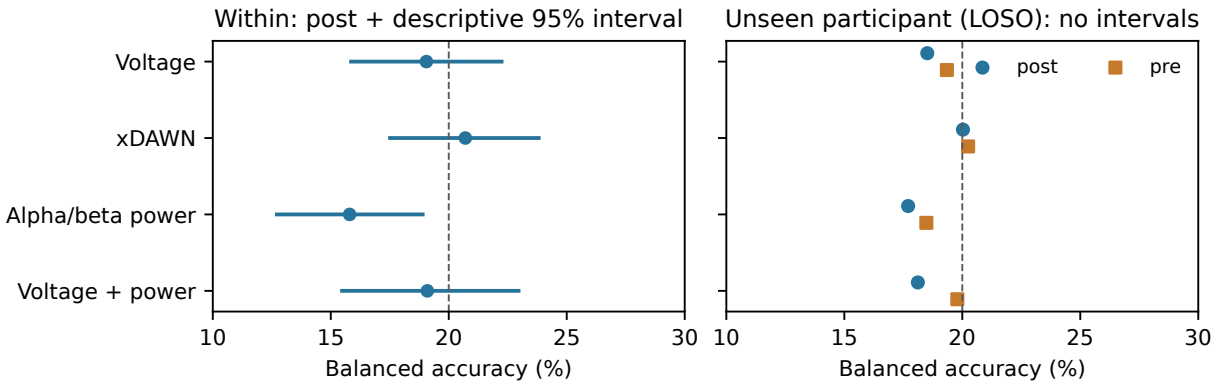


Figure 1. Post within-participant means and descriptive 95% participant-bootstrap intervals (left); pooled LOSO post/pre scores without confidence intervals (right). Dashed lines mark the 20% reference. Intervals resample participants, not epochs or overlapping folds.

The paired post improvement for xDAWN was +1.65 percentage points (descriptive interval −0.15 to +3.44); power-minus-voltage was −3.25 (−6.84 to +0.66); combined-minus-voltage was +0.04 (−2.00 to +2.06). None provides clear evidence of an overall improvement.

Restricted label permutations shifted complete pairs within participant/ comparison and optionally flipped orientation comparison-wide, retaining comparison-specific class support. xDAWN used 99 permutations: LOSO post $p = .41$ and pre $p = .59$, Holm-adjusted to .82 across two xDAWN windows. The power experiment used 199 permutations; all six voltage/power/combined LOSO tests had Holm-adjusted $p = 1$. These are separate correction families, not a single adjusted test across all experiments. Temporal nonstationarity may limit exchangeability.

Participant findings remain exploratory. At 199 permutations the 60-test participant family cannot reach adjusted .05 (minimum possible .30); this resolution limit is not evidence of participant-specific absence. There is no formal cohort-level within-participant permutation test. Near-chance pre performance also does not certify artifact-free post EEG.

4 Uncertainty and conditional sensitivity

4.1 A recorded-background recovery benchmark

A negative test needs to be interpreted in relation to what the method can detect. The conditional pilot targets the *expanded time-resolved waveform test*, not the classifiers, multiscale RMS or repeatability.

Recorded participant waveform-coefficient vectors supplied fixed background magnitudes, randomized by whole-participant signs. Thirty-two distinct sign backgrounds were shared across amplitudes, latency settings and processing branches. The maps retain each participant's structure but are not pure instrument noise; real effects can contribute to their magnitudes.

A 15-ms-σ Gaussian reference pulse, truncated at ± 60 ms, was centered at 100 or 300 ms. Between-participant latency jitter was 0 or 10-ms nominal SD, clipped at ± 2 SD; there was no trial-level jitter. The earliest possible pulse sample was +20 ms. Class amplitude multipliers were $-1, -0.5, 0, 0.5, 1$, at nominal maximum class peaks 0, 0.25, 0.5, 1, 2, 4 and 8 μV. Fixed scalp weights were C3=1, CP1=0.5, C4=−1, CP2=−0.5, others zero, preserving the scalp-average reference. This arbitrary benchmark is not a neural-source model or a claim about actual waveform physiology.

The same ideal signal was injected *after frozen preprocessing*, before differentiation/OLS, implemented by tested linear equivalence. Each alternative recomputed the exact sign null with the full location family and factor-three correction. Only mathematically unchanged locations reused exact sign-orbit maxima with the appropriate index permutation. Recovery required a corrected result within nonzero injected support, not an unrelated significant location.

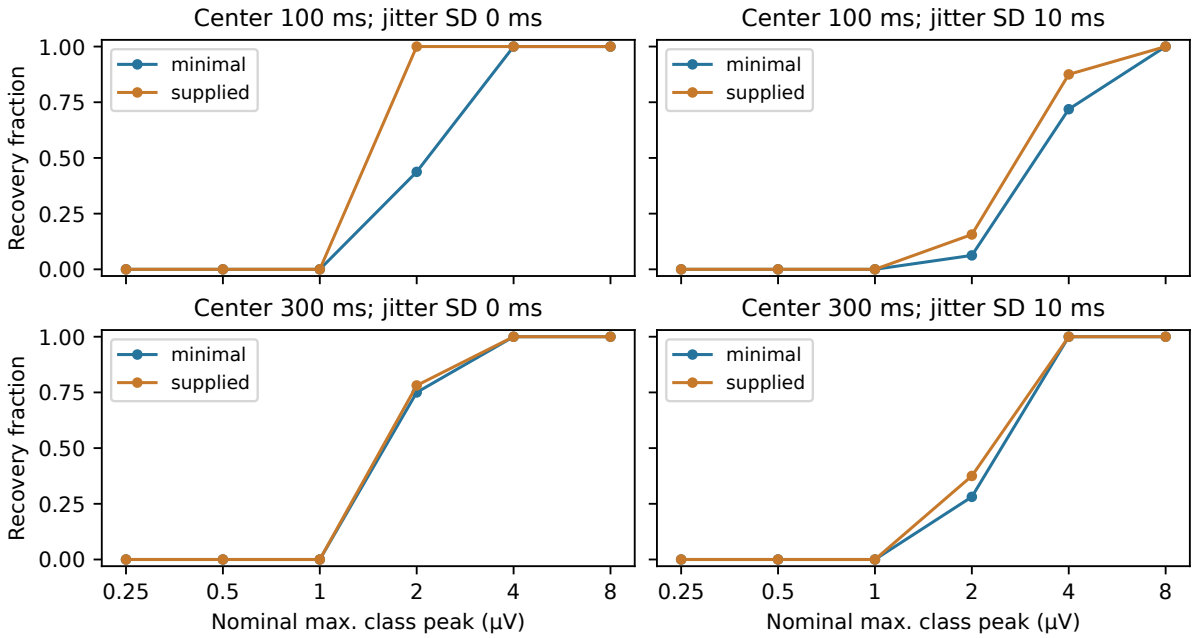


Figure 2. Conditional recovery fractions across 32 backgrounds per setting. Curves connect the fixed tested amplitudes; they do not estimate a general continuous detection threshold. Supplied labels the supplied processing branch. Nominal EEG peaks are not stimulation-current amplitudes; the maximum class-to-class peak difference is twice the stated value.

4.2 Recovery results and interpretation

Pulse center	Jitter SD	Nominal peak	Minimal recovered	Supplied recovered
100 ms	0 ms	2 μV	14/32	32/32
100 ms	10 ms	2 μV	2/32	5/32
300 ms	0 ms	2 μV	24/32	25/32
300 ms	10 ms	2 μV	9/32	12/32

Counts are out of 32 conditional replicates per setting. No 0.25, 0.5 or 1-μV injection was recovered in any setting. At 8 μV, every setting recovered the signal in 32/32 cases. At 4 μV with a 100-ms center and jitter, counts were 23/32 and 28/32; all other 4-μV settings were 32/32.

The complete zero-injection orbit enumerated all 1,024 backgrounds. An alarm in any of the three corrected families occurred in 46/1,024 (4.49%). This is a finite sign-orbit calculation, not empirical proof that real coefficient errors satisfy the required symmetry.

Meaning of the calibration. The procedure can recover stronger, consistent instances of this reference pattern; smaller injections and participant latency variation can be missed. The observed nulls therefore remain compatible with weaker or temporally variable effects. The pilot does not estimate population power, a physiological absence bound, a minimum real neural effect, or preprocessing superiority. Because injection occurs after processing, it does not quantify attenuation by ICA or filters.

Interpretation and reproducibility

- **Common versus discriminative structure:** common aligned responses were more repeatable than waveform deviations. This does not establish neural origin or guarantee single-epoch prediction.
- **Scope of negative findings:** fixed windows, features, ten participants and strict context/participant generalization constrain sensitivity. Group tests can miss individual patterns. Subsequent data-informed extensions remain exploratory; no independent replication was performed.
- **Unresolved measurement limits:** physical current timing, electrical contamination, non-charge-matched conditions, prior-stimulation carryover and recording order remain unresolved. Task 3's blocked waveform order would require separate expectation/time controls.
- **Analysis integrity:** numerical units, events, eligibility and fold identities were verified; audits and baseline replays were repeated for new representations. The follow-up suite passed 62 tests, including optimized-versus-brute-force checks and planted-signal tests. Reconstruction and source/input hashes are retained. No EEG reanalysis was performed to compile this note.

Executable analyses and reproduction commands are documented in the package READMEs under `experiments/`. Exact configurations, full corrected maps, folds, predictions and sensitivity cases remain in the original local run records. This shareable PDF contains cohort aggregates, not participant-level tables or raw recordings. Tables and figures were assembled from saved aggregate outputs; their input hashes are recorded with the ignored LaTeX project.

References

[1] R. Collu, E. J. Earley, M. Barbaro and M. Ortiz-Catalan. Non-rectangular neurostimulation waveforms elicit varied sensation quality and perceptive fields on the hand. *Scientific Reports* 13, 1588 (2023). doi:10.1038/s41598-023-28594-0.

[2] J. Whitson. *Beyond the Square Pulse: Waveform Shape, EEG Correlates, and the Pursuit of Natural Sensation in TENS*. Honors Undergraduate Thesis, University of Central Florida (2026). `https://stars.library.ucf.edu/hut2024/661/`.